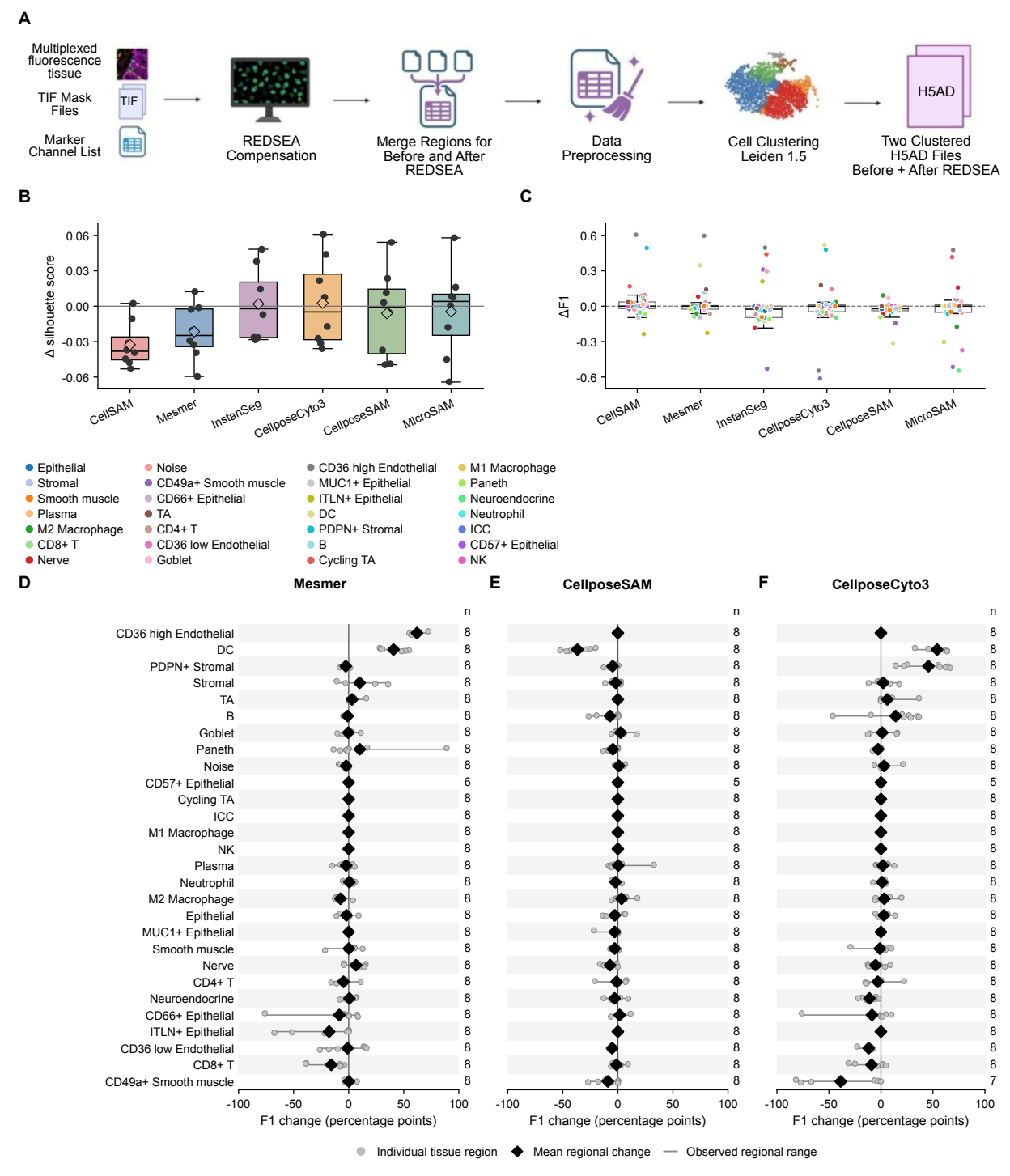

A
Multiplexed fluorescence tissue
TIF Mask Files
TIF
Marker Channel List
REDSEA Compensation
Merge Regions for Before and After REDSEA
Data Preprocessing
Cell Clustering Leiden 1.5
H5AD
Two Clustered H5AD Files Before + After REDSEA
B
Δ silhouette score
0.06
0.03
0.00
-0.03
-0.06
CellSAM
Mesmer
InstanSeg
CellposeCyto3
CellposeSAM
MicroSAM
C
ΔF1
0.6
0.3
0.0
-0.3
-0.6
CellSAM
Mesmer
InstanSeg
CellposeCyto3
CellposeSAM
MicroSAM
Epithelial
Stromal
Smooth muscle
Plasma
M2 Macrophage
CD8+ T
Nerve
Noise
CD49a+ Smooth muscle
CD66+ Epithelial
TA
CD4+ T
CD36 low Endothelial
Goblet
CD36 high Endothelial
MUC1+ Epithelial
ITLN+ Epithelial
DC
PDPN+ Stromal
B
Cycling TA
M1 Macrophage
Paneth
Neuroendocrine
Neutrophil
ICC
CD57+ Epithelial
NK
D
Mesmer
E
CellposeSAM
F
CellposeCyto3
n
CD36 high Endothelial 8
DC 8
PDPN+ Stromal 8
Stromal 8
TA 8
B 8
Goblet 8
Paneth 8
Noise 8
CD57+ Epithelial 6
Cycling TA 8
ICC 8
M1 Macrophage 8
NK 8
Plasma 8
Neutrophil 8
M2 Macrophage 8
Epithelial 8
MUC1+ Epithelial 8
Smooth muscle 8
Nerve 8
CD4+ T 8
Neuroendocrine 8
CD66+ Epithelial 8
ITLN+ Epithelial 8
CD36 low Endothelial 8
CD8+ T 8
CD49a+ Smooth muscle 8
CD57+ Epithelial 5
CD49a+ Smooth muscle 7
-100
-50
0
50
100
F1 change (percentage points)
Individual tissue region
Mean regional change
Observed regional range